

S^n , ~~S^1~~ , RP^n , $\langle \dots \rangle$
 $(n \geq 1)$
 $\downarrow$
 $\pi_1 = 0$

2-dim mfd $\underline{L(p,q)}$

$M \cup N$ $\wedge \vee$ $\langle x, s' \rangle / \langle e^{2\pi i p}, e^{2\pi i q} \rangle$
 τ_0, τ_1^2 klein.
 $a \uparrow \downarrow a$ $\text{klein. free product.}$
 $\mathbb{Z} \times \mathbb{Z}$ abababb.
 $\langle a, b \mid aba = b \rangle$
 $aba^2b^2 = e$

$ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$ $\langle a, b \mid ab = ba \rangle$ $\mathbb{Z} \times \mathbb{Z}$
 $S^1 \vee S^2$ Theorem
 $M: n\text{-dim}$ $\text{with } (n \geq 2)$
 $\pi_1(M) \cong \pi_1(M \cup \dots)$
 $\pi_1(RP^n) \cong \mathbb{Z}_2$ $(n \geq 2)$ $\pi_1(M) \cong \mathbb{Z}_2$ $\pi_1(S^n) = 0$ $(n \geq 2)$
 $\pi_1(M) \cong \pi_1(M - \{x\})$

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